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Effect of physico-chemical parameters on species biodiversity with special reference to the phytoplankton from Mushahariya chour, Madhepura, Bihar

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Abstract- A study has been conducted on physic-chemical parameters of Mushahariya chour, Madhepura, Bihar in from September 2019 to August 2021. The water quality parameters (Temperature, Dissolved oxygen, free carbondioxide, total hardness, Total alkalinity , Nitrate, Phosphate and pH) were found in the limits suggested for aquaculture practices. In the present study, 50 genera of phytoplankton were identified, of which 24 genera of Chlorophyceae, 12 genera of Bacillariophyceae, 11 genera of Cyan, two genera of Euglenophyceae and one genera of Dinophyceae. The phytoplankton abundance shows a sharp increase from January to April (pre-monsoon), with Chlorophyceae the most dominant group and Euglenophyceae as the least prevalent. A positive correlation was noticed between temperature and phytoplankton. The present study's findings show that the water quality of chour is very fertile for aquaculture practices. The results of this study will serve as baseline data for further investigations.

Key words: Physicochemical parameters, Phytoplankton, Mushahariya chour

INTRODUCTION

In addition to providing food for aquatic species, phytoplankton is crucial for preserving the biological balance and water quality. The presence of a particular phytoplankton can serve as a sign of both trophic status and water quality. The qualitative and quantitative variation related to biological productivity is comprised of physical and chemical changes in the water, and phytoplankton concentrations are sensitive to these changes. The generation of phytoplankton in the aquatic environment is influenced by the pH, dissolved oxygen, alkalinity, phosphate, nitrate, and other nutrients. Many researchers have studied the interactions between physical, chemical, and biological conditions in various aquatic habitats.[1,2] The phytoplankton community forms the base of the trophic level on which the aquatic population depends. This community is also largely influenced by the interaction of a number of physicochemical factors. The dynamic characteristics of a water body depends on colour, transparency, trophic state, and zooplankton. Fish production depends on the phytoplankton diversity and their availability. According to Davis (1955)[3], many water quality parameters, including biological factors, work simultaneously, and those factors must be considered to understand the phytoplankton population dynamics better. Karr *et al.* (2000)[4] reported that the biotic community of any water body is the product of interaction between the physicochemical parameters and geomorphological characteristics of any water body. According to Cetin & Sen (2004)[5] the distribution and variation of phytoplankton

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in freshwater lakes primarily depend on its environmental parameters. Phytoplankton communities are the primary producer and play a pivotal role in the food chain of an aquatic ecosystem.[6]

The occurrence and distribution of planktonic communities depend on several factors, viz., biotic factors, habitat physicochemical properties, and climate change.[7] The physicochemical properties of a water body played a crucial role in forming zooplankton species structure. They could also significantly impact the entire zooplankton population.

Mushahariya chour, Kherho, Madhepura, Bihar (25°55'60"N, 86°97'64"E), has been selected for this study.

MATERIAL & METHODS

The study was carried out from September 2019 to August 2021. Total ten sampling stations were chosen based on water characteristics. Water samples collected on bimonthly basis from all the sampling stations between 9.00 am to 03.00 pm. Temperature and pH were measured at the site itself. For DO estimation 250ml DO bottles were used, fixed at the site and samples brought to the laboratory to estimate DO concentration. For study of other parameters like CO_2, TA, TH, nitrate, and phosphate analysed following the APHA (2012)[8].

Plankton samples were collected using bolting silk cloth plankton net of 20 cm diameter with mesh size 20 micrometre, qualitative and quantitative analysis was performed. Samples were collected in duplicate and concentrated to 50 ml filtering 50 liters of water from the respective stations, after collecting zooplankton samples preserved in 5% formalin and phytoplankton in 4% Lugol's Iodine solution for further qualitative and quantitative analyses. Plankton was observed and identified under different magnifications using the HUND inverted microscope. Observed plankton was identified using keys and monographs.[9] Enumeration of plankton was carried out using a Sedgwick-Rafter counting cell method, the procedure outlined by Welch (1948)[10]. The average of three samples was taken into consideration, and the results are given in terms of no/litre.

RESULTS

Physiochemical parameters

The physicochemical parameters of the chour showed spatial and temporal variation in the values of the parameters following a seasonal and site-specific pattern. Detailed information of physicochemical parameters is summarised in Table 1a.and 1b. (Station-wise variation) and Table 1c. (Month-wise variation).

In the present study, air temperature ranged from 24.5 to 34.8°C. Minimum temperature (24.5°C) was recorded in January at station 1, which coincided with the winter season. Maximum temperature (34.8°C) was recorded in May at station 7, indicating the peak of the summer season. Spatially, there was an insignificant variation in water temperature. However, a time-specific variation showing an increasing trend from January to May, followed by a significant decrease up to November, was noticed. Minimum temperature (22.5°C) was recorded in the month of January at station 1, during the winter season, and maximum water temperature (31.6°C) was recorded in the month of May at station six and station seven during the peak of the summer season. A significant variation in the values of DO was also recorded, with a minimum value in January at station 1 (4.0 mg/l) and a maximum value in November at station 10 (7.8 mg/l). The results followed the general; the pattern of the inverse relation of DO on the temperature where the decrease in temperature increases the DO level due to higher solubility of oxygen at lower temperatures. The concentration of free carbon dioxide varied significantly (3.1 to 5.4 mg/l). In addition, there was a significant temporal variation in CO_2 levels during different months. Minimum free CO_2 level was recorded in November at station 8 (3.1mg/l) and in January at station 4 (5.4 mg/l).

Table 1a. Station-wise variation in the physicochemical parameters of Mushahariya chour, Madhepura, Bihar

	STN 1	STN 2	STN 3	STN 4	STN 5
Air temp. (°C)	24.5*- 29.5	24.7-31.5	25.4-31.5	26.4-32.7	26.6 -32.7
Water temp. (°C)	22.5*-27.5	22.6-29.4	23.6-29.6	25.6-30.5	25.6-30.5
Dissolved oxygen	4.0*-5.5	4.5-6.4	4.6-6.8	4.1-5.6	5.5-7.3
Carbon dioxide	3.5-5.3	3.5-5.1	3.5-4.1	3.4-5.4**	3.3-5.1
Total Alkalinity	120.66*-127.46	121.14-126.87	120.81-129.75	121.97-127.54	121.74-129.94
Total Hardness	72.40-76.84	73.27-76.71	73.49-77.94	51.48*-77.53	73.50-79.85
Nitrate-N	2.84-2.96**	1.62-1.93	1.59-1.96	1.82-2.93	1.96-2.88
Phosphate	1.82-2.14	1.76-2.09	1.77-2.53	1.82-2.32	1.77-2.92
pH (No unit)	6.4*-7.4	6.5-7.6	7.3-7.5	7.0-7.5	7.3-7.7

Table 1b. Station-wise variation in the physicochemical parameters of Mushahariya chour, Madhepura, Bihar

	STN 6	STN 7	STN 8	STN 9	STN 10
Air temp. (°C)	24.5- 29.5	24.7-34.8**	25.4-31.5	26.4-32.7	26.6 -32.7
Water temp. (°C)	25.6-31.6**	25.6-31.6**	25.4-30.7	23.5-30.7	23.4-29.4
Dissolved oxygen	4.6-7.5	5.1-7.2	4.8-7.2	4.7-6.7	6.1-7.8**
Carbon dioxide	3.3-5.3	3.4-4.5	3.1*-5.1	3.2-5.1	3.3-3.8
Total Alkalinity	122.73-130.42**	122.76-129.82	121.48-128.92	121.30-129.24	122.54-130.23
Total Hardness	72.65-79.75	72.30-80.22	72.62-79.67	72.56-79.56	74.51-80.23**
Nitrate-N	1.66-1.85	1.55-1.96	1.31*-1.86	1.60-1.95	1.60-1.95
Phosphate	1.74-2.83	1.76-3.12	1.74-2.82	1.73*-1.84	1.86-3.15**
pH (No unit)	7.3-8.3**	7.3-7.7	7.3-7.7	7.3-7.5	7.6-8.2

Table 1c. Month wise variation in the physicochemical parameters of Mushahariya chour, Madhepura, Bihar

	Sept.	Nov.	Jan.	Mar.	May
Air temp. (°C)	27.3-30.6	25.5-28.7	24.5*-27.6	25.3-29.5	29.5-34.8**
Water temp. (°C)	25.7-28.5	23.5-26.8	22.5*-24.8	23.5-27.5	27.5-31.6**
Dissolved oxygen	4.1-6.7	5.3-7.8**	4.0*-6.1	5.2-6.6	5.5-7.5
Carbon dioxide	3.4-5.3	3.1*-3.5	3.8-5.4**	3.3-4.0	3.3-3.8
Total Alkalinity	123.26-127.46	121.49-126.51	120.66*-122.73	122.31-124.45	125.44-129.23
Total Hardness	73.62-75.11	71.50-77.50	51.48*-74.51	73.11-75.63	75.65-78.65
Nitrate-N	1.33-2.96**	1.62-2.93	1.85-2.95	1.31*-2.91	1.71-2.88
Phosphate	1.75-1.93	1.74-1.96	1.73*-1.88	1.77-1.95	1.76-2.07
pH (No unit)	7.1-7.5	7.2-7.7	6.4*-7.6	7.2-7.6	7.3-7.7

(* Minimum value between the stations; ** Maximum value between the stations; Units other than mentioned are in mg/l)

The values of total alkalinity showed variations in a wide range of 120.66 to 130.42 mg/l) as $CaCO_3$ alkalinity. The minimum value was recorded at station 1 (120.66 mg/l as $CaCO_3$) whereas the maximum value was noticed at station 6 (130.42 mg/l as $CaCO_3$). Temporally, the lowest and highest values for $CaCO_3$ alkalinity was recorded in January at station 1, and maximum in July at station 6. The total hardness values varied significantly from 51.48 to 80.23 mg/l as $CaCO_3$. The minimum value of was noted at station 4 (51.48 mg/l as $CaCO_3$), and station 10 exhibited the highest value during the study months (80.23 mg/l as $CaCO_3$). Temporally, the lowest value of total hardness was recorded in January at station 4 and the maximum value in July at station 10. The concentration of nitrate-N varied from 1.31 to 2.96 mg/l. The minimum value of nitrate-N was recorded at station 8 (1.31 mg/l), and the highest value was measured at station 1 (2.96 mg/l). The values showed a temporal variation. The lowest value of nitrate-N was recorded in March at station 8 and the maximum value in September at station 1. The concentration of phosphate showed a significant variation in the range of 1.73 to 3.15 mg/l. The minimum value of phosphate was recorded at station 9 (1.73 mg/l); however, the maximum value was noticed at station 10 (3.15 mg/l). Temporally, station 9 exhibited the lowest concentration of phosphate in January. The highest value for this parameter was recorded in July at station 10. The pH value varied significantly from 6.4 to 8.3 during different sampling periods. The maximum pH value recorded was 8.3 at stations 6 and the minimum value of 6.4 was measured at station 1. Temporal variation in pH is evident from a lower value in January at station 1 and maximum in July at station 6.

Phytoplankton variations

Phytoplankton occupies the base of any aquatic trophic web and are considered as primary producers. In the present study, a total of 50 genera of phytoplankton were identified. It included 24 genera of Chlorophyceae, Bacillariophyceae, 11 genera of Cyanophyceae, two genera of Euglenophyceae and one genus of Dinophyceae.

Abundance of group

Chlorophyceae > Bacillariophyceae > Cyanophyceae > Euglenophyceae > Dinophyceae. Analysis of group-wise contribution of phyto plankton showed that Chlorophyceae formed 48%, Cyanophyceae 32%, Bacillariophyceae 15%, Euglenophyceae 3% and Dinophyceae 2% of total phytoplankton recorded.

Temporal variation in phytoplankton abundance

The phytoplankton diversity and abundance showed a seasonal pattern. The identified species of phytoplankton showed very low abundance during post-monsoon, moderate abundance during monsoon and highest during

pre-monsoon months. At the end of winter season which starts from January month, the phytoplankton number also increased. The phytoplankton abundance showed an increase from January to April (pre-monsoon) with Chlorophyceae as the most dominant group, and Euglenophyceae as the least dominant group (Table 2)

Table 2. Temporal variation in average phytoplankton abundance (no/l) from all stations

Group (no/l)	Pre monsoon (Mar -May)	Monsoon (July- Sept)	Post Monsoon (Nov -Jan)	Total
Chlorophyceae	61037	56882	61486	179405
Bacillariophyceae	20595	19503	16684	56782
Cyanophyceae	42087	41509	35614	119210
Euglenophyceae	3512	3419	3141	10072
Dinophyceae	926	2810	2416	6152

DISCUSSION

Overall water quality parameters (Temperature, DO, CO2, TH, TA, Nitrite, Nitrate, Phosphate and pH) were found in the limits suggested for aquaculture practices. According to Saini *et al.* (2015)[11], the DO level at the five sampling stations along the Narmada River near Bargi reservoir fluctuated between 3.1-6.5 mg/l and lower value recorded in the summer months. Similarly, Ubarhande (2018)[12] reported that dissolved oxygen concentration was varied from 3.5-8.95 mg/l in Vishnupuri reservoir, Nanded, Maharashtra (India). The concentration of free carbon dioxide was varied from 3.1-5.4 mg/l. Other researchers reported similar findings where the values ranged between 1.75- 5.81mg/l,[13] 3.33- 9.66 mg/l.. It was noticed that at station 1 and at station 4, the values were comparatively higher than other stations which showed visual signs of pollution with high organic load and weed infestation. The pH value fluctuated between 6.4-8.3 during different sampling periods due to inherent characteristics of the soil and geology of Mushahariya chour, Madhepura, Bihar.

In the present study, 50 genera of phytoplankton were identified. It included 24 genera of Chlorophyceae, 12 genera of Bacillariophyceae, 11 genera of Cyanophyceae, two genera of Euglenophyceae and 1 genera of Dinophyceae. According to Rasal *et al.* (2019)[14], 54 genera were identified from Bargi Dam (Narmada River), including 16 genera of Bacillariophyceae and 21 genera of Chlorophyceae 10 genera of Cyanophyceae, 2 genera by each of Chrysophyceae and Euglenophyceae, whereas three genera of Dinophyceae group. Sharma *et al.* (2011)[15] also reported that a total of 14 genera belonging to Bacillariophyceae (5 genera), Cyanophyceae (8 genera) and Euglenophyceae (1 genera) at Dograwadaghat in river Narmada.

The phytoplankton abundance shows a sharp increase from January to April (pre-monsoon), with Chlorophyceae as the most dominant group and Euglenophyceae as the least prevalent. This variation in phytoplankton number (high in summer and low in winter) may be due to high temperatures. A positive correlation was noticed between temperature and phytoplankton. According to Khare (2005)[16], there was a significant correlation between plankton density and temperature, DO, phosphate, and nitrate. Several researchers have suggested that water temperature plays a pivotal role in algal growth.[17]

CONCLUSION

The present study's data shows that the water quality of the selected chour is good for aquaculture practices except for a few stations. But its water quality deteriorating due to high human interferences, pushing this ecosystem towards the process of eutrophication. The data and information presented in this paper will serve as baseline data for further investigations.

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